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The Source of Innovation: Boundary Spanner

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ABSTRACT *There appears to be a consensus that organizational integration across functional and disciplinary specialties drives superior firm capabilities. The basic assumption is that new practices and concepts emerge from the interaction of individuals engaged in different functional departments. However, communication across functional boundaries is often difficult. As such, Tushman & Scanlan suggested that functional boundaries can be spanned effectively only by boundary spanners who understand the coding schemes that are attuned to the contextual information on both sides of the boundary. The study of boundary spanners has significant performance implications to firms. Compared to its importance, theoretical development and empirical inquiries into this phenomenon are relatively scant. Thus, this study explores the factors that determine who becomes a boundary spanner and examines boundary spanners' performance implications. A Taiwanese company provided the context for our study. Partial Least Squares (PLS) was used to test the theoretical model. The results imply that a company needs to have an appropriate mixture of boundary spanners and non-boundary spanners, because boundary spanners have higher innovative performance.*

KEY WORDS: Boundary spanners, PLS, social network analysis

Introduction

Scholars have long been concerned about the search for the various characteristics that affect organizational performance. The research has ranged from the industrial organization view (Porter, 1985) developed in the 1980s to the resource-based view (RBV) of the firm developed in the 1990s (Barney, 1991; Wernerfelt, 1984). However, sophisticated customer demands, shorter product lifecycles and new market opportunities are forcing organizations and academics to search for the dynamic capabilities rather than focusing on a static industry structure or resource endowments. Because firms compete on dynamic capabilities, it has become very important to understand the nature and origins of these dynamic capabilities. There appears to be a consensus that organizational

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integration across functional and disciplinary specialties drives superior firm capabilities. Studies of product development also conclude that differences in firms' performance can be attributed largely to the degree of cross-boundary knowledge integration in their development process (Bond *et al.*, 2004; Griffin & Hauser, 1996; Sherman *et al.*, 2000; Song *et al.*, 1997). Flatter hierarchies and cross-functional teams aim at improving communication among different departments. The basic assumption is that new practices and concepts emerge from the interaction of different functional departments. That is, the interaction of individuals engaged in different functional departments is one of the sources of dynamic capabilities.

However, communication across functional departments is often difficult and prone to bias, because each department has its own specialized semantics. A loss of information is almost inevitable because it is necessary to recode at the boundaries between departments. With each recoding, the richness in the meaning of words (e.g. rules or conventions) in one context is lost when translating them into another context. Accordingly, Tushman and Scanlan (1981a) suggested that functional departments can be spanned effectively only by boundary spanners who understand the coding schemes that are attuned to the contextual information on both sides of the boundary. This enables boundary spanners to search out relevant information on one side and disseminate it on the other. Because the study of boundary spanners has significant performance implications to firms, insights into the factors that determine who becomes a boundary spanner are very important.

Compared to its importance, theoretical development and empirical inquiries into this phenomenon are relatively scant. To fill this void, this study explores the factors that determine who becomes a boundary spanner and examines boundary spanners' performance implications. A Taiwanese company provided the context for our study. The results imply that a company needs to have an appropriate mixture of boundary spanners and non-boundary spanners, because boundary spanners would have normal job performance (they would not be the best employee, just a 'good enough' employee) but may have higher innovative performance.

The organization of this study is as follows. In the next section, we propose the theoretical model. In the third section, how samples are derived and the definitions of measures are given. In the fourth section, we analyze the results. Finally, we conclude with a discussion and future research directions.

Theory and Hypotheses

When firms are facing a radically changing environment, they must respond to uncertain conditions by finding non-routine ways to mobilize diverse skills, to coordinate activities, and to allow for different opinions to increase organizational flexibility and innovation capabilities. Given the opportunity to access resources across formal boundary lines, boundary spanners can help blur the boundaries of formal function units and stimulate communication between departments so that they can easily reconfigure and mobilize various resources in response to emerging threats or opportunities (Galbraith, 1973; Gittel, 2002; Tushman & Nadler, 1978). With an understanding of how one department would affect other departments, boundary spanners could further explore the root causes of problems, challenge assumptions and then develop novel solutions. As a result, this research implies that a company needs to have an appropriate number of boundary spanners. Figure 1 encapsulates our theoretical model.

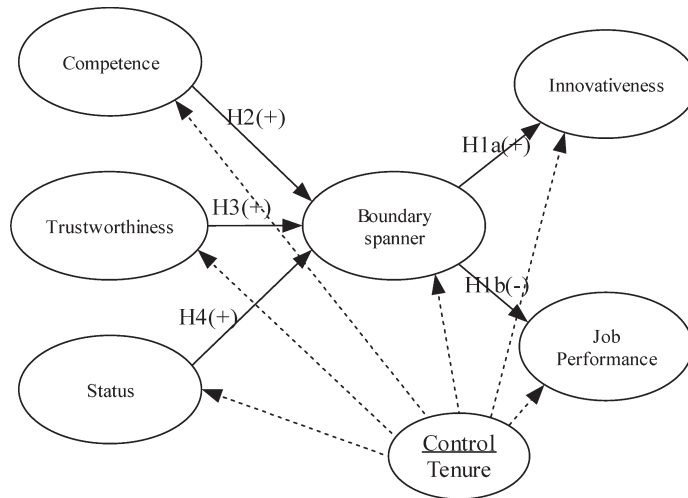


Figure 1. Theoretical model

Performance Consequences of Boundary Spanner

As organizations evolve, specialized units are created to deal with relatively homogeneous tasks. Employees are trained and socialized to attend to the units' particular work requirements (Brown & Duguid, 2001). For example, R&D professionals are hired to maintain and develop technology; marketing specialists are hired to sell the product, talk to customers and communicate product benefits. Over time these groups grow apart, each expert at their own function, but less aware of the other's contribution (Griffin & Hauser, 1996). Functional boundaries are therefore developed, separating specialized units from each other. This specialization is a double-edged sword, for it facilitates transmission of tacit knowledge (Cross *et al.*, 2001), simplifies coordination (Ancona & Caldwell, 1992), increases learning rates (Levinthal & March, 1993) and avoids potential conflicts (Pelled *et al.*, 1999). However, such specialization simultaneously creates obstacles to information processing between the specialized unit and external areas (Griffin & Hauser, 1996; Tushman & Scanlan, 1981a). The specialization of organization members turns organizations into distributed knowledge systems in which the range of knowledge required for innovation is dispersed over organizations (Tsouksa, 1996).

In this context, a boundary spanner's ties provide a channel to access diverse information, knowledge or resources across formal boundary lines. Early access to different perspectives and information is important in providing a basis for innovative ideas. In the product innovation literature, the general view is that project effectiveness can be attributed largely to the timely integration of knowledge across functional boundaries (Bond *et al.*, 2004; Griffin & Hauser, 1996; Sherman *et al.*, 2000; Song *et al.*, 1997). Moreover, researchers have claimed that innovative ideas often spring up at the boundaries of specialties by bringing together disparate knowledge (Brailsford, 2001; Granovetter, 1973; Kanter, 1988; Leonard-Barton, 1995; Nahapiet & Ghoshal, 1998). In fact, what seems a misunderstanding for one discipline may be a new twist for another discipline and the active resolution of this misunderstanding may lead to a new jointly created

knowledge – creative ideas. The key to creative thought appears to be the combination and reorganization of different thinking and concepts to advance new understandings.

These accumulating evidences suggest that individuals with ties across different departments gain non-redundant information concerning opportunities and resources sooner than those without such ties. Their ties serve as channels for knowledge or information exchanges, enabling them to have more chances to come up with creative ideas. Previous research suggested that knowledge within one clique might be largely redundant, providing little additional information over what an individual might already know (Granovetter, 1973, 1985). Thus, actors whose ties are limited to one department are less likely to receive diverse information (impeding their ability to develop new ideas) than those whose ties span multiple departments. Thus, we hypothesize:

Hypothesis 1a: A boundary spanner is more innovative than those who are not boundary spanners.

We suggest that a boundary spanner is likely to have a lower level of job-related performance. The benefits of external relationships may be offset by the costs involved in maintaining such relationships. Because maintaining external linkages are very time consuming, boundary spanners might run short of the time and resources necessary for their department's specific work tasks. Thus, boundary spanners may have difficulties in completing their tasks on time and not achieving the highest performance ratings in their department. Since they spend less time in their unit's particular work, their expertise might be lower than those who are not boundary spanners. Thus, we hypothesize:

Hypothesis 1b: A boundary spanner achieves a lower level of job-related performance than those who are not boundary spanners.

Antecedents for Boundary Spanner

Competence. To understand the idiosyncratic norms and languages of different departments, boundary spanners require a range of intellectual capabilities from the various departments with which they connect (Cross & Prusak, 2002). Their intellectual capabilities help them appreciate what seems a conflict between departments, but in reality is an opportunity for both. Accordingly, one of the possible reasons that few marketing managers can become boundary spanners between marketing and R&D is that they lack the technological background to appreciate the opportunities identified by R&D. Research therefore suggested that perceived work-related competence is a primary determinant for boundary spanners (Tushman & Scanlan, 1981b). Employees who are perceived by their colleagues as technically competent are more likely to become boundary spanners. Moreover, as boundary spanners take time to consult with and advise individuals from different departments, they evolve to understand the idiosyncrasies between different departments (e.g. non-verbal signs, values or languages), which in turn increases their perceived competence. Thus, we hypothesize:

Hypothesis 2: An employee's work-related competence is positively associated with his or her odds of becoming a boundary spanner.

Trustworthiness. Trust has often been seen as an informal self-enforcing agreement alternative to formal governance mechanisms (Dyer & Singh, 1998; Yli-tenkari *et al.*, 2001). It reduces the probability for opportunism and the need for costly monitoring mechanisms, encouraging cooperative behavior and knowledge exchange. As a result, trust promotes the exchange of a range of specific assets, such as knowledge, information and supports that are difficult to put a price on (Kale *et al.*, 2000; McEvily *et al.*, 2003; Uzzi, 1997).

As an employee gains a reputation of being dependable and trustworthy, other actors begin to treat this reputation as an important signal that will influence their preferences in selecting partners. Accordingly, people in different departments are more likely to approach an employee whom they perceive as trustworthy when they have problems or opportunities. This provides the trustworthy employees with more chances to become boundary spanners. Moreover, those who are helped by the trustworthy employee might feel obligated to give him a privileged 'favor', such as providing critical information or offering necessary support even with rush jobs. This further increases his chances to establish new relationships, access new opportunities and gain timely support from different departments. Accordingly, we hypothesize:

Hypothesis 3: An employee's trustworthiness is positively associated with his or her odds of becoming a boundary spanner.

Status (Formal Rank). Several researchers have found that differences in formal rank are likely to influence patterns of interaction in organizations (Cohen & Zhou, 1991; Saunderson *et al.*, 1994). High-status individuals are likely to be valued as important by others and are treated more tolerantly. Conversely, those low in status can sometimes be ignored even if their input is intelligent and creative. Moreover, higher-status individuals, by virtue of their control over resources and their decision-making authority, may be better positioned to emerge as boundary spanners (Tushman & Scanlan, 1981a) because people are more willing to communicate with them regarding their work opportunities or problems. As a result, high-status individuals are more likely to become boundary spanners. We hypothesize:

Hypothesis 4: An employee with high level of status is positively associated with his or her odds of becoming a boundary spanner.

Methods

Research Site and Data Collection

This research was undertaken at CL (a pseudonym) company. Founded in 1995, CL Corp. is a world leader and pioneer in bringing advanced digital video and audio software to the market. CL also delivers solutions that include video editing and recording, online training and distance learning solutions, digital media presentation and delivery, and unique talking-head design tools for the web, using advanced MPEG-4 technologies. With rapid growth and consistent profitability, CL enjoyed a record-breaking IPO in October of 2000 on the Taiwan Over The Counter Exchange. To keep up with market demands,

CL has operations in North America, Europe and the Asia Pacific region including Japan. The company founder emphasized the importance of innovation and creativity as the keys to survival in this competitive marketplace. The CL organizational structure was deliberately kept flat, with only four levels of hierarchy. It has 12 functional units: R&D, marketing, human resource, administration, finance, information management, legal, product commercialization, product planning, quality assurance, sales and customer support. Because of the potential opportunities for knowledge to be fertilized across different departments, we chose CL as the sample company.

In April 2004, the survey instrument was sent to each of the CL employees, asking them to complete the questionnaire. At the time, the CL employed 200 people in Taiwan alone. The questionnaire was designed to collect both relational and non-relational data. We obtained relational data by asking the respondents to specify their relations with other employees. Non-relational data were gathered mostly through questions using Likert-type scales. The performance-rating data were collected using a separate questionnaire sent to all 21 supervisors.

Because social network research is one of the most intrusive types of organization research, there is undoubtedly a general mistrust of any attempt to gain data about how one interacts with others and their job performance. As we relied on supervisors to rate employee performance, this created more concerns in both supervisors and the company. Firms are unwilling to divulge such information without strong assurances that the information provided will in no way harm them, yet this same information could provide insight in how to improve their organization.

After a personal visit to the company to explain the purpose of the research, we gained full support from the CEO of CL. Due to the potential damage to the firm that could occur if these data are leaked, we reached an agreement that all employee names would be transformed into code numbers (e.g. 0408) before the data were released to us. This code number was performed by only one person, designated by the company. Throughout the data collection process, individuals – including supervisors – were assured that their responses would be kept confidential and that all results would be presented only on an aggregate level. The response rate was 62% for the questionnaire to all employees and 59% for the questionnaire sent only to supervisors.

Measures

Boundary spanner. To assess the network relations, we follow the work of Sparrowe (2001) by asking the respondents one question: ‘Who do you go to for help or advice on work-related matters?’ Respondents were free to nominate as many as five contacts. After coding the data into the socio-matrices measuring social interaction, we calculated the in-degree centrality for each individual concerning to intra- and inter-department contacts. The in-degree centrality of an actor (individual i) was the number of individuals that ask individual i work-related matters in their answers to the above question. Consistent with Tushman & Scanlan (1981b), internal communication stars were defined as those individuals in the top third of the intra-department communication and external communication stars were defined as those who are in the top third of an external communication distribution. And boundary spanners are those who are both internal and external communication stars (Tushman & Scanlan, 1981b).

Trust. We developed two questions to examine the inter-personal trusting relationships in the company: (1) 'Please indicate whom you can rely on without any fear that they will take advantage of you' and (2) 'In general, who will always keep the promises they make to you?' To measure an actor's trustworthiness, we calculated the in-degree centrality for each individual to measure the level of trustworthiness in his answers to the above question.

Competence. Following the work of Tushman & Scanlan (1981a), we used one question to assess the level of intellectual capability of individuals. The item was: 'Please indicate who are the most competent persons in the company.' To create a single-item measure of competence, we calculated the in-degree centrality for each individual in their answers to the above question.

Status. There are four levels of hierarchy in the company. From company records, we coded rank as 0 for non-supervisors, 1 for supervisors and 2 for senior supervisors who manage some supervisors.

Performance measures. In the absence of objective performance measures of employees within the company, we relied on supervisory ratings. That is, the job performance measured here emphasizes the extent to which individuals succeed (in the eyes of management) in contributing to organizational ends. Using a six-item scale arranged in five-point Likert format, supervisors rated the performance of those subordinates who reported directly to them. We informed supervisors that performance ratings would be confidential and used only for research purposes. Following prior research (Mehra *et al.*, 2001), supervisors related subordinates' innovativeness on three items: (1) the degree to which the subordinate generated creative work-related ideas; (2) the degree to which the subordinate promoted and championed work-related ideas to others; and (3) the degree to which the subordinate sought new technologies, processes, techniques and/or product related ideas. Supervisors related subordinates' performance on three items: (1) the overall job performance of the individual; (2) the likelihood that the subordinate would achieve future career-related success (such as promotions, awards, bonuses, and involvement in high-profile projects); and (3) the likelihood you would pick the subordinate to succeed you in your job.

Control variables. Tenure was included as control variables because prior work has suggested that the length of time a person has been with the company is likely to affect his or her work performance.

Results

Performance Implications of Boundary Spanners

The research model was tested using PLS, a structural modeling technique that is well suited for highly complex predictive models (Wold, 1985). PLS has several strengths that made it appropriate for this study, including its ability to measure theoretical models during the early stage of development where the theoretical model and its measures are not well formed, and its ability to measure small sample

size requirements¹ (Barclay *et al.*, 1995). Since all the constructs have reflective indicators in the theoretical model, the minimum sample size is determined by the largest number of antecedent constructs leading on to endogenous constructs. The sample size is 124, exceeding the recommended minimum of 40. In PLS, item reliability is assessed by examining loadings (see Table 1). All loadings exceeded the recommended threshold of 0.7 (Carmines & Zeller, 1979).

To assess the convergent validity of constructs, researchers using PLS report the internal consistency measure developed by Fornell & Larcker (1981), which is similar to Cronbach's alpha. Fornell & Larcker argued that their measure is more appropriate to Cronbach's alpha since it uses the item loadings obtained with the causal model. Table 1 lists the internal consistency of each construct. Nunnally (1978) suggested that 0.7 as a benchmark for mode reliability was applicable in the early stages of research. The internal consistency of all constructs was larger than 0.7, except for the construct of innovativeness.

To assess discriminant validity, Fornell & Larcker (1981) suggested the use of Average Variance Extracted (AVE), which should be greater than the variances shared between the construct. The comparison can be made in the correlation matrix (see Table 2), including the correlations between different constructs in the off-diagonal elements of the matrix, and the square roots of the AVE for each of the constructs along the diagonal. For adequate discriminant validity, the diagonal elements should be greater than the off-diagonal elements in the corresponding rows and columns. Table 2 indicates that all constructs had adequate discriminant validity.

Table 1. Survey items

<i>Competence (Internal consistency = 1)</i>	Loading
Please indicate who are the most competent persons in the company (in-degree centrality)	1.00
<i>Trustworthiness (Internal consistency = 0.80)</i>	
Please indicate whom you can rely on without any fear that they will take advantage of you (in-degree centrality)	0.88
In general, who will always keep the promises they make to you? (in-degree centrality)	0.91
<i>Boundary spanner (Internal consistency = 1)</i>	
Who do you go for help or advice on work-related matters? (in-degree centrality)	1.00
<i>Innovativeness (Internal consistency = 0.65)</i>	
The degree to which the subordinate generated creative work-related ideas	0.89
The degree to which the subordinate promoted and championed work-related ideas to other	0.71
The degree to which the subordinate searched out new technologies, processes, techniques and/or product related ideas	0.80
<i>Job performance (Internal consistency = 0.77)</i>	
The overall job performance of the individual	0.91
The likelihood that the subordinate would achieve future career related success	0.93
The likelihood you would pick the subordinate to succeed you in your job	0.79

Table 2. Correlations and discriminant validity

Latent variables	1	2	3	4	5	6	7
1. Competence	1.00						
2. Trustworthiness	0.24	0.94					
3. Status	0.40	0.24	1.00				
4. Boundary spanner	0.30	0.21	0.39	1.00			
5. Innovativeness	0.15	0.12	0.21	0.31	0.92		
6. Performance	0.11	0.11	0.13	0.07	0.66	0.95	
7. Tenure	0.06	0.26	0.14	0.14	0.04	0.14	1.00

Structural Model

Having established confidence in our measurement model, we examined the main effects. The test of the structural equation model includes estimation of the path coefficients and R^2 values. The path coefficients indicate the strengths of the relationships between the dependent and independent variables and R^2 values represent the amount of variance explained by the independent variables. Using LVPLS version 1.8 (Lohmoller, 1981), we determine path coefficients. To further test the path significance, we employed the Jack-knife methods.

Table 3 reports the results of hypothesis testing, the amounts of explained variance in the constructs of the boundary spanner, innovativeness and job performance. Three of the five hypotheses were supported. A boundary spanner was more innovative than someone who was not a boundary spanner ($b = 0.31, p < 0.001$). Therefore, Hypothesis 1a was supported. A boundary spanner did not achieve a lower level of job-related performance than those who were not boundary spanners ($b = 0.05, p$ n.s.). Therefore, Hypothesis 1b was not supported. An individual's work-related competence had a significant positive effect on his odds for becoming a boundary spanner ($b = 0.25, p < 0.01$).

Table 3. Path estimates and variance explained

Hypothesis	Path	Path estimate
H1a	Boundary spanner (BS) → Innovativeness	0.31***
H1b	BS → Job performance	0.05
H2	Competence → BS	0.25**
H3	Trustworthiness → BS	0.08
H4	Status → BS	0.30**
Control	Tenure → Innovativeness	0.01
Control	Tenure → Job performance	0.13
Control	Tenure → BS	0.07
Control	Tenure → Competence	0.06
Control	Tenure → Trustworthiness	0.26**
Control	Tenure → Status	0.13 +
Construct		R^2 value
Innovativeness		0.19
Job performance		0.12
BS		0.29

+ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Thus, Hypothesis 2 was supported. On the other hand, an individual's trustworthiness had no significant effect on his chances for becoming a boundary spanner ($b = 0.08$, p n.s.); although the coefficient for trustworthiness was positive, as predicted. Therefore, Hypothesis 3 was not supported. An individual's status was found to be positively associated with his odds for becoming a boundary spanner ($b = 0.30$, $p < 0.01$). Hypothesis 4 was therefore confirmed.

The R^2 value represents the percentage of variance in an endogenous construct explained by other constructs connected to it directly. To evaluate the model, the R^2 value was reported in Table 3. Interpreted like multiple regression results, the R^2 value indicates the amount of variance explained by the model. The overall model explained 19% of the variance in innovativeness and 12% of the variance in job-related performance. In addition, the model explained 29% of the variance in the boundary spanner construct.

Discussions and Conclusions

The study explored the factors that lead to someone becoming a boundary spanner, and examined the performance consequences of a boundary spanner. The results indicate that boundary spanners are more innovative than non-boundary spanners (H1a) because individuals with ties across different departments gain non-redundant information sooner than those without such ties. On the other hand, our findings suggest that boundary spanners did not achieve a lower level of job-related performance than non-boundary spanners. There were no job-related performance differences between them (H1b). These results indicate that a company needs to have an appropriate mixture of boundary spanners and non-boundary spanners, because boundary spanners would have normal job performance but may have higher innovative performance. The results are consistent with previous findings (Teigland & Wasko, 2003), which suggested that interactions between individuals at the same location tend to have a negative effect on their creativity, while individual interactions between different locations have a positive effect on creativity.

Our results suggest that the antecedents for boundary spanners are competence (H2) and status (H4). Put differently, if an individual is perceived to be competent and has a higher status, he or she is more likely to become a boundary spanner. Previous research suggested that perceived work-related competence is a primary determinant of boundary spanners (Tushman & Scanlan, 1981b). The intellectual capabilities of boundary spanners help them appreciate the conflicts between different departments. In terms of status, higher-status individuals, by virtue of their control over resources and their decision-making authority, are better positioned to emerge as boundary spanners.

However, contrary to our expectation in Hypothesis 3, an individual's trustworthiness had no significant effect on their chances of becoming a boundary spanner. One potential explanation for the result is that people become boundary spanners because others contact them for job completion or technical problems. Thus, whether a boundary spanner is trustworthy is not very important because the main purpose of the contact is for solving problems.

Implications for Practice

As organizations evolve, specialized units are created to deal with specialized tasks. However, such specialization reduces the probabilities for resource recombination

between different departments. As a fundamental requirement for knowledge creation is to draw upon different knowledge, this study suggests that boundary spanners who connect with a number of otherwise disconnected departments provide themselves with the initial conditions for innovation. Such access, as Kanter (1988) observed, 'allows innovators to go across formal lines and levels in the organization to find what they need.' A company can reap substantial benefits by recognizing its boundary spanners because of their networks of relationships and intellectual capabilities, constituting a valuable resource for the organization. An organization without boundary spanners tends to have reduced innovation by limiting the awareness of opportunity, alternative approaches and the perspectives of others.

Corporate life may not be friendly to boundary spanners, who might spend too much time developing an external network and less time in the immediate work. Supervisors may view their broad boundary-spanning interests as unfocused or undisciplined (Davenport & Prusak, 1998: 30). If boundary spanners are evaluated primarily on their job performance, they will have less incentive to do boundary-spanning jobs. Furthermore, boundary spanners cannot easily be replaced because relationships are not built overnight. People need a certain amount of time to develop a sufficient understanding of the jargon, needs, and requirements of the departments with which they connect. As a result, firms should encourage boundary spanners by providing resources and tangible or intangible incentives, enabling them to perform the task of relationship management across department boundaries. As the development of cross-boundary contacts can be time-consuming and may divert attention from performing productive activities, it is preferable to have boundary spanners do all the networking instead of having everyone engage in it.

This study also provides another important insight for organizations by recognizing the antecedents for becoming boundary spanners. For example, an organization can actively arrange for someone to be a boundary spanner if they have the intellectual capabilities and know-how to mobilize resources embedded within different functional units. A new managerial job would take responsibility for boundary spanning by getting the right people together from different departments to develop opportunities.

Note

1. PLS requires a minimum sample size that equals 10 times the greater of (1) the indicators on the most complex formative construct or (2) the largest number of antecedent constructs leading an endogenous construct.

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